Mars Habitat

Campaign 1 - Case 03 - Arcadia Planitia, Mars

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Teacher quick start

Students act as Data Analysts. They compare spectral distribution with an adequate total PPFD reading, reject “not enough total light,” diagnose selective red/deep-red rejection by the collector filter, and write CER.

At a glance	Case 03 - Mars Habitat
Duration	60 minutes; gameplay target 18-20 minutes
Correct diagnosis	Light-delivery filtering removes red wavelengths needed in the game chlorophyll-biosynthesis mechanism.
Core evidence	280 PPFD; 12 m silica light pipe; 68% aggregate transmission; filter replaced 47 sols ago; 92%, 88%, 31%, and 12% band transmission; FS-7 required and BP-4 incorrect.
Collect	Student Mission or selected Tasks 3, 5, 7, and 9.
Fallback	Use the controlled evidence digest; keep all tasks and rubric unchanged.

60-minute timeline

Time	Teacher move	Student action
0-5	Launch the adequate-light/yellow-growth contradiction.	Read mission question; predict needed evidence.
5-10	Frame quantity versus distribution without revealing diagnosis.	Complete Task 1.
10-28	Circulate; redirect to sensors, plants, and archive.	Play and collect targeted evidence.
28-38	Check exact values and no invented data.	Complete Tasks 2-4.
38-50	Prompt mechanism and alternative rejection.	Complete Tasks 5-7.
50-60	Transfer and exit ticket.	Complete Tasks 8-9.

Likely sticking point: “Adequate” describes total photon quantity, not complete spectral adequacy. Ask: Adequate total of what, distributed how?

Formal lesson plan: outcomes and standards

Overview

Students investigate a Mars potato-growth failure by distinguishing a scalar quantity measurement from spectral distribution. They use direct game-provided transmission values, the collector-filter record, tissue pattern, and controlled conditions to reject alternatives and explain the case mechanism.

Measurable learning objectives

- Interpret PPFD as photon quantity over a defined waveband rather than a complete spectrum.
- Graph and compare wavelength-band transmission without inventing missing measurements.
- Use symptom location and timing to infer a failure in new pigment formation.
- Reject an alternative that conflicts with adequate total intensity and healthy roots.
- Write a concise Claim-Evidence-Reasoning explanation.

Standards alignment

- NGSS MS-LS1-6: construct a scientific explanation based on evidence for the role of photosynthesis.
- NGSS MS-PS4-2: model reflection, absorption, and transmission of waves through materials.
- Practices: analyzing and interpreting data; constructing explanations; argument from evidence.
- Crosscutting concepts: cause and effect; systems and system models; energy and matter.

Success criteria

- Uses 280 umol m-2 s-1 and all four exact transmission values.
- Explains that adequate total photon quantity does not guarantee adequate spectral distribution.
- Identifies red at 31% as the lowest-transmission band within 400-700 nm.
- Explains why new tissue is affected before older tissue.
- Avoids “more brightness always fixes it” and “plants use only red” claims.

Formal lesson plan: preparation and procedure

Materials and preparation

- Game or technical fallback digest; role-appropriate mission packet; pencil.
- Open Case 03 and confirm audio/headphone expectations.
- Print at 100% / Actual Size. Keep the Answer Key private.
- Do not preview the correct diagnosis or describe red light as the only useful wavelength.

Procedure

Phase	Teacher actions	Student evidence product
Launch	Present the contradiction: an adequate PPFD reading with bleaching new growth. Ask what one total number cannot reveal.	Task 1: measurement limits.
Gameplay	Prompt students to revisit crew, sensors, plants, logs, and deep archive rather than relying on symptoms alone.	Runtime notes and exact values.
Data checkpoint	At Task 2, require only the four provided values. At Task 3, require explicit rejection of low total PPFD.	Transmission comparison and quantity-quality explanation.
Diagnosis	At Task 5, require one selected and one rejected diagnosis. At Task 6, require a complete causal chain.	Diagnosis and mechanism model.
CER/transfer	Require data-specific evidence and a next measurement, not a generic “add brighter lights” fix.	Tasks 7-9.

Checks for understanding

- Can the student explain why two lights with the same PPFD can have different spectra?
- Can the student identify red at 31% as the lowest transmission inside 400-700 nm?
- Does the student separate the game mechanism from broader red/blue/green plant responses?

Teacher case analysis

Story problem

Potato plants grew normally for about three weeks. Then top-canopy and newest leaves became pale yellow to white while older lower leaves retained green. Roots remained healthy, and iron/nitrogen additions did not help.

Evidence channels

CREW

Top-down progression; nutrient additions failed.

SENSORS

12 m light pipes plus white LEDs; 280 PPFD.

PLANTS

Healthy roots; pigment failure concentrated in new tissue.

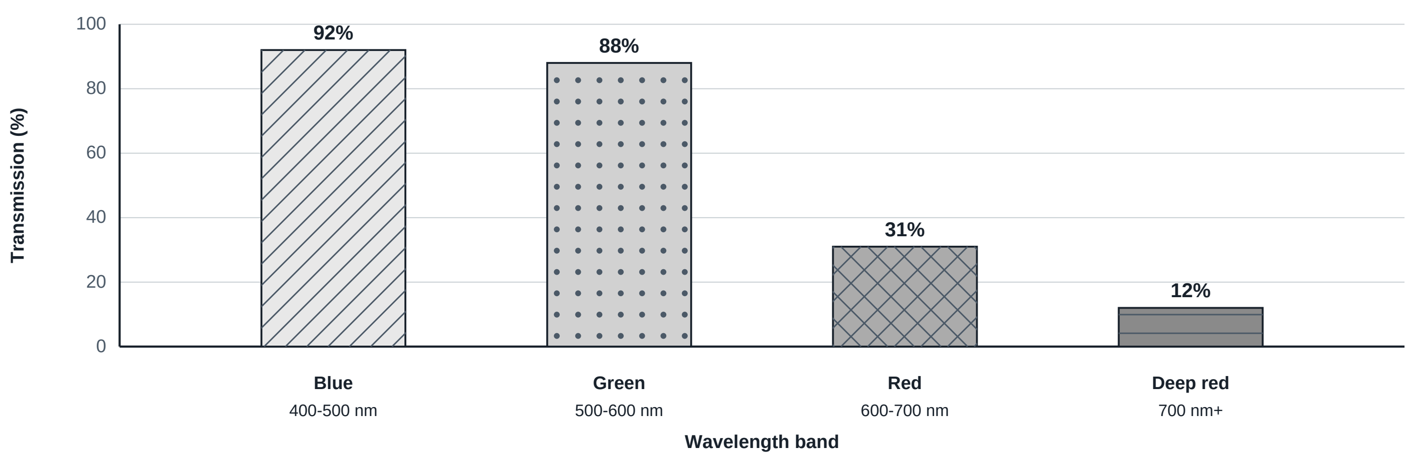
LOGS / ARCHIVE

Filter replaced 47 sols ago without a logged part number; FS-7 required and BP-4 incorrect.

Essential versus supporting evidence

Evidence	Instructional weight
280 PPFD described as adequate	Essential: rejects simple low total light.
31% red transmission; 12% deep-red transmission	Essential: reveals selective rejection.
New tissue bleaches first	Essential: points to new chlorophyll formation.
Healthy roots; nutrients failed	Essential for rejecting root/nutrient alternatives.
20 C, CO2 1200 ppm, UV index 0.1	Supporting controls; do not overinterpret.

Surface intake to habitat output transmission



SOURCE STATUS: Numerical bars are exact game values; general plant-light principles come from primary research.

Mechanism, distractors, and misconceptions

Correct case mechanism

47-sol filter replacement → **BP-4 rejects red/deep red** → **case-critical red delivery low** → **new chlorophyll development impaired** → **pale new growth**

Case mechanism model. General plant-light responses also involve blue, green, and red wavelengths.

Within the game model, the collector/light-pipe system delivers enough total photons for an adequate PPFD value while a likely BP-4 filter disproportionately rejects red/deep-red wavelengths. New tissue must form new chlorophyll; the game links the light-dependent POR step to that process. The pattern therefore appears first in new growth.

Why the wrong answers are tempting

Option	Why tempting	Why it fails
Perchlorates poison roots	Mars setting activates prior knowledge.	Roots are healthy; crop uses a prepared nutrient system.
CO_2 too high	1200 ppm sounds extreme.	Runtime states it is within tolerance and symptoms do not fit.
Mars sol disrupts photoperiod	Mars day length is memorable.	Does not explain wavelength data or new-tissue pattern.
Not enough total light	Bleaching suggests weak light.	PPFD is 280 and described as adequate.

Misconceptions to prevent

- PPFD is a complete description of light quality.
- A brighter lamp always fixes a lighting failure.
- Green photons are useless.
- Only red light matters to every plant process.
- Yellowing automatically proves nutrient deficiency.

Assessment, rubrics, and supports

Quick rubric

Dimension	Secure	Developing	Beginning
Data use	Uses 280 and all four exact transmission values.	Minor omission or error.	Omits or invents values.
Quantity vs spectrum	Clear distinction and rejection.	Partial distinction.	Treats PPFD as a complete spectrum.
Mechanism	Complete data-to-symptom chain.	Cause identified; chain incomplete.	Conclusion without mechanism.
Alternatives	Rejects with evidence.	Weak rejection.	No evaluation.
Communication	Clear CER and vocabulary.	Understandable but incomplete.	Major gaps.

Formal analytic rubric

- Accurate game data
- Graph interpretation and quantitative comparison
- Quantity-versus-distribution distinction
- Mechanistic reasoning
- Competing-diagnosis evaluation
- Source-status and uncertainty awareness
- Scientific communication and completion

Differentiation and accessibility

- Use the six-page Accessible Mission with larger type and linear tables.
- Permit oral dictation, speech-to-text, or typed response.
- Read source-status labels aloud.
- No arithmetic is required; students compare direct measurements.
- Provide the technical fallback without reducing reasoning demand.

Academic grading boundary

Assess evidence, mechanism, reasoning, vocabulary, alternative evaluation, and completion. Do not grade game score, speed, rapport, optional dialogue, or clue-discovery order.

References and source-status guidance

Authoritative and primary sources

- Illuminating Engineering Society: PPFD definition as photon quantity per area and time in the 400-700 nm band.
- McCree (1971/1972): wavelength-dependent action spectra across 22 crop species with broad blue and red maxima.
- Hogewoning et al. (2010): equal irradiance with different blue:red composition produced different plant function.
- Heyes, Ruban, and Hunter (2003): POR is a light-driven chlorophyll-biosynthesis enzyme.
- Armstrong et al. (1995): light-dependent protochlorophyllide reduction in angiosperm chlorophyll biosynthesis.
- Bula et al. (NASA/HortScience, 1991/1994): controlled-environment LED sources provide specified PAR flux and spectral regions.

Game-specific and curriculum-original data

Item	Status
280 PPFD; 12 m silica pipe; aggregate transmission 68%	Game-provided Case 03 runtime data.
92%, 88%, 31%, 12% transmission	Direct game-provided band readings.
47-sol replacement; FS-7 and BP-4 filter models	Game-specific maintenance and hardware records.
Mars symptoms and diagnosis	Game-specific controlled case model.
Graphs and mechanism diagrams	Curriculum-original vector constructions.

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Technical fallback and release gate

Fallback principle

Technical failure must not become academic failure. Give students the same controlled evidence, exact tasks, and rubric. Do not provide the diagnosis.

Controlled evidence digest

CREW

Three weeks of normal growth; newest leaves become washed out; iron and nitrogen do not help.

SENSORS

12 m light pipe plus white LEDs; combined PPFD 280; no spectral analysis had been performed.

PLANTS

Healthy roots; older leaves retain green; new growth is pale yellow-white.

ARCHIVE

Filter replaced 47 sols ago without a logged part number. Required FS-7; incorrect BP-4 passes blue/green and rejects most red/deep red. Transmission: 92%, 88%, 31%, 12%.

Facilitation

- Read the digest once and allow students to revisit it.
- Require Tasks 1-9 unchanged.
- Use the same quick or formal rubric.
- Allow speech-to-text or oral response as documented accommodations; arithmetic is not required.

Physical-print gate

Automated static, browser, PDF, portability, accessibility, overflow, checksum, and rendered-review checks may pass while the release remains unreleased. Owner physical 100%-scale printing must still verify clipping, line survival, patterns, direct labels, and writing usability before approval.

Owner print checks

- Print each role at 100% / Actual Size on ordinary classroom equipment.
- Confirm compact page identity and role-specific N of total footers.
- Confirm graph axes, units, data labels, and patterns survive black-and-white copying.
- Confirm response boxes match expected answer length and bottom reserve remains intentional.
- Confirm no production metadata appears on ordinary printable pages.